Matrix Spaces; Rank 1; Small World Graphs

Vectors don't have to be columns of numbers. Whole matrices can be vectors too, and a covariance matrix lives in one such space. Then the simplest matrices of all, rank one, which is what a single-factor market looks like. And a first look at graphs.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **Conditions and important concepts**
   12 short definitions: matrix spaces, symmetric / upper / diagonal subspaces, rank one matrices, graphs.
2. **Procedure**
   Find the dimension of a matrix space; test for rank one and split it into uv^T.
3. **2 × 2 examples**
   A two-asset covariance matrix; a one-factor return table.
4. **3 × 3 examples**
   A three-asset covariance matrix; one factor (rank 1) and two factors (rank 2).
5. **Bigger example: the market model**
   Covariance $= \sigma^2 \beta\beta^T + D$. 1,275 numbers become 101. With a factor builder.
6. **Small world graphs**
   A broker network: distances, the adjacency matrix and two-step links.
7. **Final memory table**
   Everything on one screen.

METHODS IN THIS LESSON: Matrix spaces | Symmetric matrices S | dim(S + U) formula | Rank one: uv^T | Single-factor model | Graphs and distance | Adjacency matrix

PART 2 Prerequisites

Click any link to review it, then come back.

- Basis and dimension (Lesson 9) and the four subspaces (Lesson 10).
- Symmetric matrices and covariance (Lesson 5).

- Columns times rows (Lesson 3): a column times a row is a whole matrix.

PART 3

Explanation

THE NUMBERS USED IN THE EXAMPLES

Betas (how much a fund moves when the market moves 1%): Nifty 1, midcap 1.2, balanced 0.5, bonds 0. **Market returns** in three scenarios (bad, normal, good year): -10%, 5%, 20%. **Market variance** $\sigma^2 = 400$ (a 20% volatility, since $20^2 = 400$).

1. Conditions and important concepts

1. Vectors don't have to be columns

A vector space only needs two moves: **add** two of its things and **scale** one by a number, without leaving the space (Lesson 5). Matrices can do that. So can functions. So:

$$\boxed{\text{Anything you can add and scale can be a "vector".}}$$

2. The space M of all 3 × 3 matrices

DEFINITION · MATRIX SPACE M

M = all 3×3 matrices. Adding two gives a 3×3 matrix; scaling gives one; the zero matrix is in it. It has **dimension 9**: you choose 9 numbers.

A basis: the 9 matrices with a single 1 and zeros elsewhere, $E_{11}, E_{12}, \ldots, E_{33}$ (E_{ij} has its 1 in row i, column j). Every matrix is $\sum a_{ij} E_{ij}$, and no E_{ij} is a mix of the others. M behaves just like $\mathbb{R}^9$.

3. Symmetric matrices S: where covariance lives

DEFINITION · THE SUBSPACE S OF SYMMETRIC MATRICES

S = all 3×3 matrices with $A^T = A$. Sums and multiples of symmetric matrices are symmetric, so S is a subspace. **Dimension 6**: choose the 3 diagonal entries and the 3 entries above the diagonal; the 3 below are copies.

A covariance matrix is symmetric (Lesson 5), so it lives in S. For n assets you need

$$\boxed{\dim S = \frac{n(n+1)}{2}} \qquad n = 3: \ 6 \qquad n = 50: \ \frac{50 \times 51}{2} = 1275$$

4. Upper triangular matrices U

All 3×3 matrices with zeros below the diagonal. A subspace of dimension 6 (3 diagonal + 3 above). Basis: $E_{11}, E_{12}, E_{13}, E_{22}, E_{23}, E_{33}$.

5. Diagonal matrices D = S ∩ U

Matrices that are both symmetric and upper triangular must have zeros off the diagonal. Dimension 3. A diagonal covariance matrix means the assets are **uncorrelated**: only their own variances, no covariances.

6. Union vs sum

$S \cup U$ ("symmetric **or** upper triangular") is **not** a subspace: adding a symmetric matrix to an upper triangular one usually gives neither. The **sum** $S + U$ (all $s + u$) is a subspace, and here it is all of M.

> **RULE · DIMENSIONS OF A SUM AND AN INTERSECTION**
>
> $$\dim S + \dim U = \dim(S + U) + \dim(S \cap U) \qquad 6 + 6 = 9 + 3 \ \checkmark$$

7. Rank one matrices

> **RULE · EVERY RANK ONE MATRIX IS UVᵀ**
>
> A matrix has rank 1 exactly when it is a column times a row: $A = uv^T$. Every column is a multiple of u; every row is a multiple of v^T.

$$\begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \end{bmatrix} \qquad \text{(MIT's example)}$$

8. Spotting rank one

Eliminate: if only **one** non-zero row survives, the rank is 1. Equivalently, every row is a multiple of the first non-zero row. For a 2×2: $ad - bc = 0$ (and not all zero).

9. Rank one in markets: one factor

> **MODEL · SINGLE-FACTOR RETURNS**
>
> If every fund's return is its beta times the market's return, then the table of returns (funds × scenarios) is $\beta\, r_m^T$: a column of betas times a row of market returns. **One factor drives everything ⇒ the table has rank 1.**

10. Rank one matrices are building blocks

A matrix of rank r can be written as a sum of r rank-one matrices. In markets: an r-factor model. But the sum of two rank-one matrices is usually rank 2, so the set of rank-one matrices is **not** a subspace (and neither is "all rank 4 matrices", MIT's example).

$$\begin{bmatrix}1\\0\end{bmatrix}\begin{bmatrix}1 & 0\end{bmatrix}+\begin{bmatrix}0\\1\end{bmatrix}\begin{bmatrix}0 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\qquad \text{rank } 1+\text{rank } 1=\text{rank } 2$$

11. A subspace from a rule: zero-cost trades with four funds

All (v_1, v_2, v_3, v_4) with $v_1+v_2+v_3+v_4=0$ is the null space of $A=\begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}$. Rank 1, so its dimension is $4-1=3$, with basis $(-1,1,0,0)$, $(-1,0,1,0)$, $(-1,0,0,1)$: switch Nifty into each other fund. The other three subspaces: $C(A)=\mathbb{R}^1$, row space of dimension 1, left null space $\{0\}$.

12. Graphs

> **DEFINITION · GRAPH AND DISTANCE**
>
> A **graph** is a set of **nodes** joined by **edges**. The **distance** between two nodes is the smallest number of edges on a path between them. A "small world" graph has short distances even with many nodes ("six degrees of separation").

2. Procedure

A. Dimension of a matrix space

Step	What to do
1	Check it's a subspace: the zero matrix is in it, and sums and multiples stay in it.
2	Count how many entries you can choose freely; the rest are forced.
3	Basis: one matrix per free choice (a 1 in that spot, plus whatever copies it forces).
4	Dimension = number of basis matrices.

B. Is it rank one? Split it into uv^T

Step	What to do
1	Take the first non-zero row: that's v^T.
2	For each row, find the number that turns v^T into it: those numbers form u.
3	If some row isn't a multiple of v^T, the rank is more than 1.
4	Check $uv^T = A$ entry by entry.

3. 2 × 2 examples

Example A: a two-asset covariance matrix (dimension 3)

Nifty variance 400, bonds variance 25, covariance 30:

$$\text{Cov} = \begin{bmatrix} 400 & 30 \\ 30 & 25 \end{bmatrix} = 400 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + 25 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + 30 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Check entry by entry: top-left $400 + 0 + 0 = 400$ ✓, bottom-right $0 + 25 + 0 = 25$ ✓, both off-diagonal $0 + 0 + 30 = 30$ ✓. The three matrices on the right are a basis of the 2×2 symmetric matrices, so that space has **dimension** $3 = \frac{2 \times 3}{2}$: two variances and one covariance.

Example B: a one-factor return table (rank 1)

Nifty (beta 1) and midcap (beta 1.2), market return -10% in a bad year and 20% in a good year:

$$\begin{bmatrix}1\\1.2\end{bmatrix}\begin{bmatrix}-10 & 20\end{bmatrix}=\begin{bmatrix}1(-10) & 1(20)\\1.2(-10) & 1.2(20)\end{bmatrix}=\begin{bmatrix}-10 & 20\\-12 & 24\end{bmatrix}$$

- Elimination: row 2 $-\ 1.2\times$ row 1 $=(-12+12,\ 24-24)=(0,0)$. One non-zero row: **rank 1**.
- $ad-bc=(-10)(24)-(20)(-12)=-240+240=0$ ✓.
- Meaning: midcap's row is just $1.2\times$ Nifty's row. One factor (the market) explains the whole table.

4. 3 × 3 examples

Example A: a three-asset covariance matrix (dimension 6)

Nifty, midcap, bonds:

$$\text{Cov}=\begin{bmatrix}400 & 480 & 0\\480 & 900 & 0\\0 & 0 & 25\end{bmatrix}$$

Six numbers decide it: the variances 400, 900, 25 and the covariances 480 (Nifty–midcap), 0 (Nifty–bonds), 0 (midcap–bonds). A basis of S: E_{11}, E_{22}, E_{33}, $E_{12}+E_{21}$, $E_{13}+E_{31}$, $E_{23}+E_{32}$. So

$$\text{Cov}=400E_{11}+900E_{22}+25E_{33}+480(E_{12}+E_{21})+0(E_{13}+E_{31})+0(E_{23}+E_{32}).$$

Example B: one factor, three funds (rank 1)

$$R=\beta\, r_m^T=\begin{bmatrix}1\\1.2\\0.5\end{bmatrix}\begin{bmatrix}-10 & 5 & 20\end{bmatrix}=\begin{bmatrix}-10 & 5 & 20\\-12 & 6 & 24\\-5 & 2.5 & 10\end{bmatrix}\begin{matrix}\leftarrow\text{Nifty}\\\leftarrow\text{midcap}\\\leftarrow\text{balanced}\end{matrix}$$

- Row 2 $-\ 1.2\times$ row 1: $(-12+12,\ 6-6,\ 24-24)=(0,0,0)$.
- Row 3 $-\ 0.5\times$ row 1: $(-5+5,\ 2.5-2.5,\ 10-10)=(0,0,0)$.
- One non-zero row left: **rank 1**. Every column is a multiple of $\beta=(1,1.2,0.5)$; every row is a multiple of $r_m^T=(-10,5,20)$.

Example C: add a second factor (rank 2)

Suppose midcap also has a small-cap factor that adds $(4,-2,6)$ to its returns. That's a second rank-one piece:

$$R_2 = \begin{bmatrix} 1 \\ 1.2 \\ 0.5 \end{bmatrix} \begin{bmatrix} -10 & 5 & 20 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 4 & -2 & 6 \end{bmatrix} = \begin{bmatrix} -10 & 5 & 20 \\ -8 & 4 & 30 \\ -5 & 2.5 & 10 \end{bmatrix}$$

Midcap's row: $(-12+4,\ 6-2,\ 24+6) = (-8, 4, 30)$. Is it a multiple of Nifty's row? $-8/-10 = 0.8$, $4/5 = 0.8$, but $30/20 = 1.5$. No. Balanced is still $0.5\times$ Nifty. So **rank 2**: two factors, two rank-one pieces.

5. Bigger example: the market model

Where did Example A's covariance matrix come from? From a **single-factor model**: each fund's risk is (its beta × the market's risk) plus its own private risk.

$$\text{Cov} = \sigma^2\, \beta\beta^T + D \qquad \beta = \begin{bmatrix} 1 \\ 1.2 \\ 0 \end{bmatrix} \text{ (Nifty, midcap, bonds)}, \quad \sigma^2 = 400$$

Step 1, the rank-one market part:

$$400\, \beta\beta^T = 400 \begin{bmatrix} 1\cdot 1 & 1\cdot 1.2 & 1\cdot 0 \\ 1.2\cdot 1 & 1.2\cdot 1.2 & 1.2\cdot 0 \\ 0 & 0 & 0 \end{bmatrix} = 400 \begin{bmatrix} 1 & 1.2 & 0 \\ 1.2 & 1.44 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 400 & 480 & 0 \\ 480 & 576 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 2, the private (diagonal) part: Nifty 0 (it *is* the market), midcap 324, bonds 25:

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 324 & 0 \\ 0 & 0 & 25 \end{bmatrix} \qquad \text{Cov} = \begin{bmatrix} 400 & 480 & 0 \\ 480 & 576+324 & 0 \\ 0 & 0 & 25 \end{bmatrix} = \begin{bmatrix} 400 & 480 & 0 \\ 480 & 900 & 0 \\ 0 & 0 & 25 \end{bmatrix}$$

Why it matters. A full covariance matrix for 50 stocks lives in S, with $\dim = 1275$ numbers to estimate. The market model needs only 50 betas + 1 market variance + 50 private variances $= 101$. A rank-one piece plus a diagonal is a huge simplification, and it's how risk models start.

$$\boxed{1275 \text{ numbers (full covariance)} \ \rightarrow\ 101 \text{ numbers (one factor)}}$$

Try it: build a factor table

Betas: Nifty 1 Midcap 1.2 Balanced 0.5

Market return (%): bad -10 normal 5 good 20

☐ Add a small-cap factor to midcap: $(4, -2, 6)$

Return table (rows: Nifty, midcap, balanced; columns: bad, normal, good):

$$\beta r_m^T = \begin{bmatrix} -10 & 5 & 20 \\ -12 & 6 & 24 \\ -5 & 2.5 & 10 \end{bmatrix}$$

Rank 1: one factor explains the whole table (every row is a multiple of (-10, 5, 20)).

6. Small world graphs

Six brokers A–F. An edge means two brokers trade directly with each other:

Edges	A–B, A–C, B–C, B–D, C–D, D–E, E–F

METHOD · ADJACENCY MATRIX

Put a 1 in row i, column j when nodes i and j are joined, 0 otherwise. It is symmetric. Its square counts **two-step** paths: entry (i, j) of A^2 = number of ways to go from i to j in exactly two steps.

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \qquad A^2 = \begin{bmatrix} 2 & 1 & 1 & 2 & 0 & 0 \\ 1 & 3 & 2 & 1 & 1 & 0 \\ 1 & 2 & 3 & 1 & 1 & 0 \\ 2 & 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

- $(A^2)_{AD} = 2$: A reaches D in two steps two ways, A–B–D and A–C–D. Check: row A $(0, 1, 1, 0, 0, 0)$ · column D $(0, 1, 1, 0, 1, 0) = 0 + 1 + 1 + 0 + 0 + 0 = 2$ ✓.
- Diagonal of A^2 = number of trading partners: B has 3 (A, C, D) ✓.
- Distances from A: B 1, C 1, D 2, E 3, F 4. The farthest pair (A and F) is 4 steps apart.

Why a trader cares. In a financial network, distance is how many hops a shock takes to travel. "Small world" networks, where any two banks are only a few hops apart, spread trouble fast. Lesson 12 turns graphs into matrices properly, with directions on the edges.

7. Final memory table

Concept	Key fact	Example
Matrix space M (3×3)	dimension 9; basis E_{ij}	behaves like $\mathbb{R}^9$
Symmetric S	dimension $n(n+1)/2$	covariance: 6 numbers for 3 assets, 1275 for 50
Upper triangular U	dimension 6 (for 3×3)	U from elimination
Diagonal $D = S \cap U$	dimension 3	uncorrelated assets
Sum rule	$\dim S + \dim U = \dim(S + U) + \dim(S \cap U)$	$6 + 6 = 9 + 3$
Rank one	$A = uv^T$; every row a multiple of one row	one-factor returns βr_m^T
Rank r	a sum of r rank-one pieces	an r-factor model
Not subspaces	$S \cup U$; all rank-1 matrices	rank 1 + rank 1 = rank 2
Graph	nodes + edges; distance = fewest edges	broker network, A to F = 4
Adjacency A	$(A^2)_{ij}$ counts two-step paths	A to D: 2 ways

▼ **Check yourself (MIT problem 11.2): with $A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ acting on 3×3 matrices X, which X give $AX = 0$, and what are the dimensions?**

A has rank 2 and $N(A)$ is the multiples of $(1, 1, 1)$. $AX = 0$ means every column of X is in $N(A)$, so $X = \begin{bmatrix} a & b & c \\ a & b & c \\ a & b & c \end{bmatrix}$: dimension 3. The matrices AX are those whose columns each add to 0: dimension 6. And $3 + 6 = 9 = \dim M$.

▼ **Check yourself (MIT recitation): the 2 × 3 matrices whose null space contains $(2, 1, 1)$. A subspace? Dimension?**

Yes: if $A(2,1,1) = 0$ and $B(2,1,1) = 0$, then $(A+B)(2,1,1) = 0$ and $(cA)(2,1,1) = 0$. Each row (r_1, r_2, r_3) must satisfy $2r_1 + r_2 + r_3 = 0$: 2 free numbers per row, 2 rows, so **dimension 4.**